## Chapter 6, Problem 38P

### Problem

The current through a 40-mH inductor is

$$i(t) = \begin{cases} 0, & t < 0 \\ te^{-2t}\ \text{A}, & t > 0 \end{cases}$$

Find the voltage $v(t)$.

### Step-by-step solution

**Step 1** of 1

$$V = L\frac{di}{dt} = 40\times10^{-3}\left(e^{-2t} - 2te^{-2t}\right)dt$$

$$= 40\left(1-2t\right)e^{-2t}\ mV,\ t > 0$$